

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 07)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

.....

.....

.....

.....

.....

.....

0 1 . 2

Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

.....

.....

.....

.....

.....

.....

0 1 . 3

Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

.....

.....

.....

0 1 . 4

Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

.....

10

0 2 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1

2

0 2 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

0 2 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

5

03 . 1

Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking

.....

.....

.....

DNA databases and eDNA

.....

.....

.....

Image recognition software

.....

.....

.....

Acoustic monitoring / sonograms

.....

.....

.....

03 . 2

Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

.....

.....

.....

.....

10

0 4

The Isle of Eigg (area 31 km², population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

0 4

. 1

Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4

. 2

Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

1

2

3

0 4

. 3

Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

[2 marks]

.....

.....

.....

0 4 . 4

Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

10

0 5 . 1

Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

0 5 . 2

Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

05 . 3

Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

05 . 4

Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

10

06

The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

06 . 1

Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer thousand tonnes

0 6 . 2 Calculate the maximum catch that would have kept the stock constant. **[1 mark]**

Answer thousand tonnes

0 6 . 3 Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult. **[3 marks]**

.....

.....

.....

.....

.....

.....

.....

0 6 . 4 Describe how each of the following measures obtained from commercial catches can indicate overfishing. **[4 marks]**

Catch per unit fishing effort

.....

.....

.....

Mean fish size and mean age

.....

.....

.....

10

07 . 1

Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

.....

.....

.....

.....

07 . 2

Explain how soil organisms provide services that humans depend on.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

07 . 3

Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

.....

.....

07 . 4

Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

10

08

Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

08 . 1

Use Table 1 to explain which site is more suitable for maize.

[3 marks]

0 8 . 2

Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

1

.....

2

.....

3

.....

0 8 . 3

Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

.....

.....

.....

.....

0 8 . 4

Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

.....

.....

.....

.....

10

09

Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

09**1**

Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer

%

09**2**

Explain why so little of the solar energy is stored in the crop.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

09**3**

Explain why few food chains have more than four trophic levels.

[2 marks]

.....

.....

.....

.....

09**4**

Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

09 . 5

Explain how pigs can increase the efficiency of a food production system.

[2 marks]

10

10 . 1

Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

1	0
---	---

2

[1 mark]

Do not write outside the box

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a full page of a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

Do not write outside the box

END OF QUESTIONS